Engineering Compact, Affinity-Tuned De Novo Mini-Protein Binders for Next-Generation Cell Therapy Integration

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Antigen Recognition in Engineered Cell Therapy

Chimeric antigen receptor (CAR) and other engineered immune cell therapies rely mainly on antibody-derived scFv domains for antigen recognition[1]

Limitations of scFv's

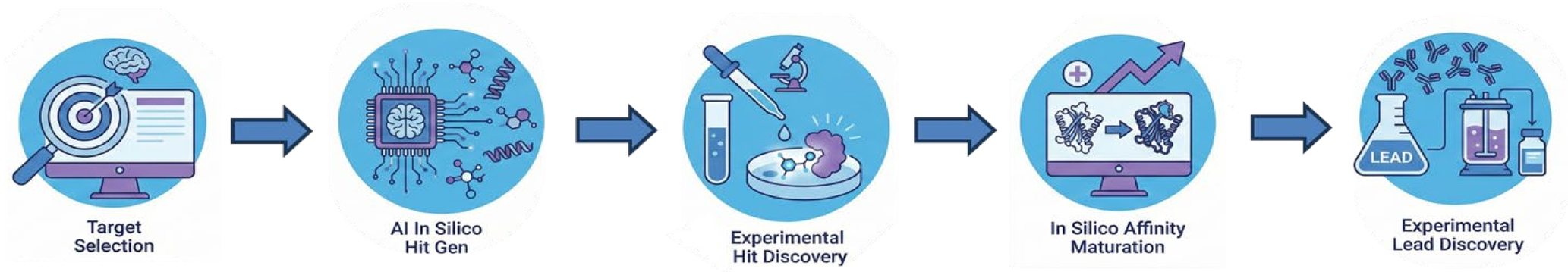
While clinically validated, scFvs present constraints:

- Large multi-domain architecture (~250 aa)
- Aggregation propensity and stability challenges
- Tonic signalling risk associated with surface properties
- Inflexibility for multi-specific and logic-enabled receptors

Opportunity - De novo protein binders (DNPB)[2]

- Small single domain(50–80 aa) [1]
- Structurally defined, monomeric scaffolds
- Robust and tuneable physicochemical properties
- Reduced extracellular footprint
- Potential compatibility with multi-specific receptor systems

De novo protein binder platform



Target Selection	AI In Silico Hit Gen	Experimental Hit Discovery	In Silico Affinity Maturation	Experimental Lead Discovery
Choice of epitope, hit generation feasibility assessment. Project plan.	>20 binders passing filtering criteria	Cloning, expression (*E. coli*) & purification. Target binding validation and Kd determination.	>20 sequences identified	Cloning, expression (*E. coli*) & purification. Target binding validation and Kd determination.

Designed mini-proteins demonstrate:

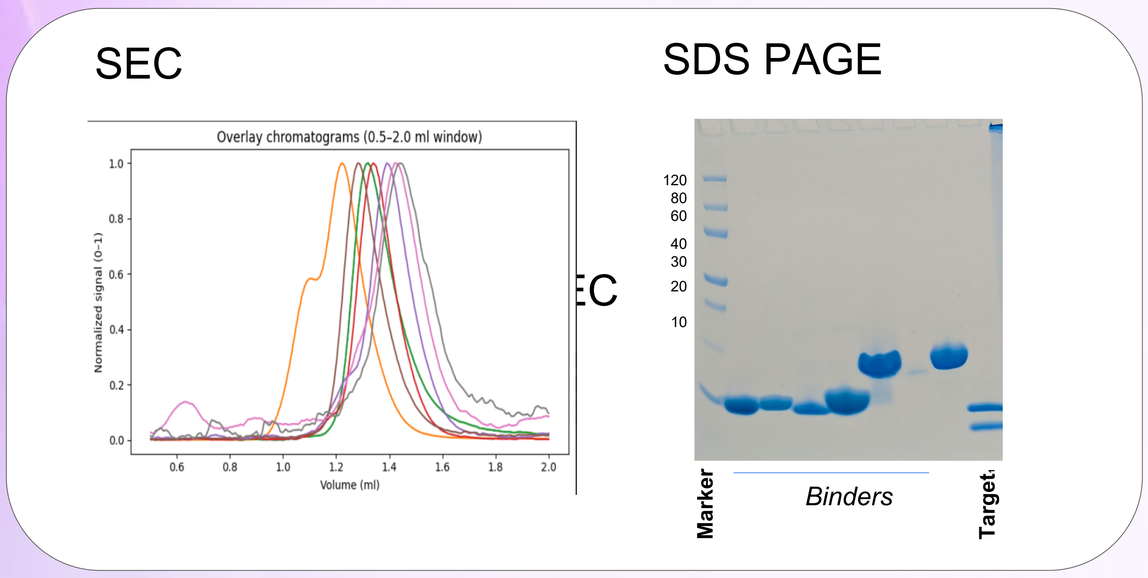
- Robust soluble expression in bacterial systems
- Monodisperse profiles by size-exclusion chromatography
- High purity by SDS-PAGE
- High thermal stability - assessed by DSF
- High affinity target binding, after affinity maturation cycles

Affinity Tuning for Cell Therapy Applications

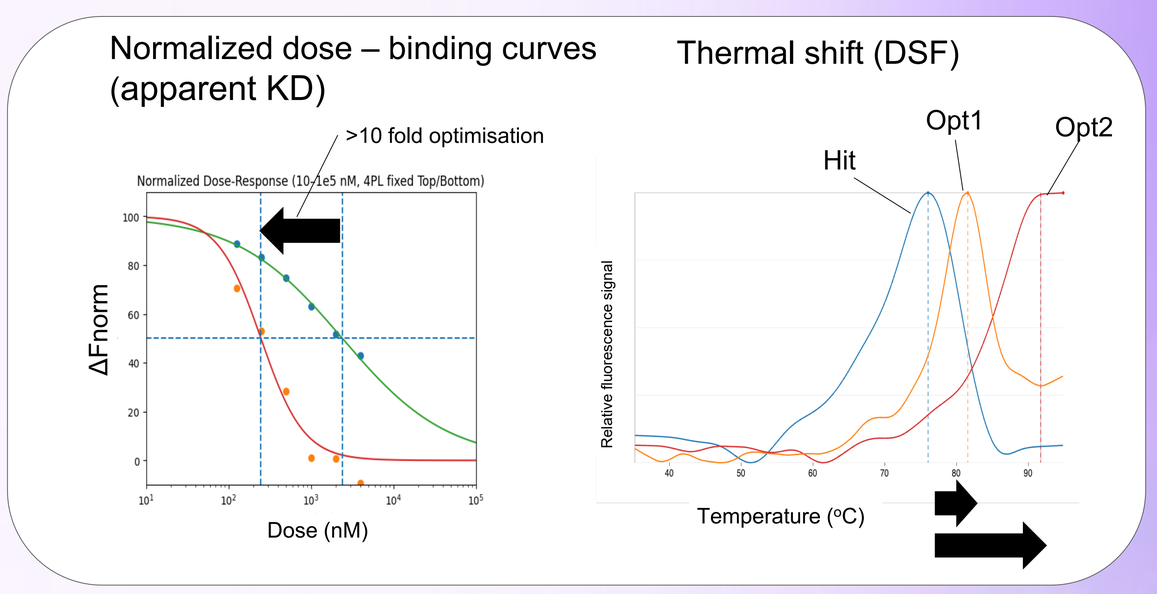
CAR performance depends on antigen recognition and affinity tuning (~10–100 nM) to achieve selective activation.
Our platform enables:

- Systematic affinity maturation
- Generation of binder panels spanning defined KD ranges
- Control of physicochemical properties
- Design for CAR integration

Protein production



Affinity maturation



Validated Binder Discovery Package (10-12 weeks)

- Initial target feasibility assessment - *free*
- Deep-dive analysis, design and plan
- Data package:
  - Binding affinity measurements (KD)
  - SEC mono-dispersity profiles
  - Thermal stability data (DSF)
  - SDS-PAGE purity analysis & mass spec QC
  - Mini-protein binder panel sequences (<5)
- Samples: protein (0.1-1 mg) and expression vector DNA
- *Optional: further affinity maturation cycle(s)*

Outlook

DNPB are promising modular antigen-recognition components for engineered immune cell systems.

- Alternative recognition domains for CAR constructs
- Multi-specific or tandem receptor formats
- Logic-enabled receptor architectures

Would you like to evaluate integration of compact de novo binders within your next engineered cell therapy platform?

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References:

1. **Nix et al.,** 2024). *Alternative target recognition elements for chimeric antigen receptor (CAR) T cells:* **Cytotherapy**, 26, 729–738.
2. **Chow et al.,** *Sequence and structural determinants of efficacious de novo chimeric antigen receptors.* **bioRxiv preprint.**
3. **Watson et al**. (2023) *De novo design of protein structure and function with RFdiffusion.* **Nature**, 620, 1089–1100.

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